SPETRO GEOPORTAL ACADEMY | OFFICIAL COURSE HANDBOOK

PostGIS Spatial Database Management & Spatial SQL

Author: Regina Obe & Leo Hsu

Scope: Practical Field Manual & Reference Guide | Pages: 10

Course Overview & Objectives

Practical Spatial SQL handbook covering geometries, spatial indexing (GIST/SP-GIST), ST_Buffer, ST_Intersection, routing, and raster SQL querying.

MODULE OUTLINE:

- **1. Getting Started with PostGIS**
- **2. Spatial Indexing & Query Optimization**
- **3. Spatial Relationships & Predicates**
- **4. Geometric Transformations & Geoprocessing**
- **5. Serving Spatial Data to the Web**

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1. Getting Started with PostGIS

Enabling the extension, spatial types (GEOMETRY and GEOGRAPHY), SRID management, and spatial table schema conventions.

KEY TAKEAWAYS & PRACTICAL LAB INSTRUCTIONS

1. Load corresponding dataset layers from the SPETRO Data Harvester or Hugging Face.
2. Execute geoprocessing algorithms and verify outputs in the Interactive Map Editor.
3. Export cartographic layouts, GeoTIFF rasters, or GeoJSON vector polygons for field use.

2. Spatial Indexing & Query Optimization

GiST (Generalized Search Tree) indexes, bounding box operator (&&), ST_DWithin vs. ST_Distance performance tuning, and explain analyze benchmarking.

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3. Spatial Relationships & Predicates

DE-9IM matrix foundations, ST_Contains, ST_Intersects, ST_Within, ST_Crosses, and ST_Touches.

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4. Geometric Transformations & Geoprocessing

ST_Buffer, ST_Intersection, ST_Union, ST_Difference, ST_Centroid, ST_ConvexHull, and Voronoi diagram generation.

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5. Serving Spatial Data to the Web

ST_AsGeoJSON, ST_AsMVT (Mapbox Vector Tiles generation inside SQL), pg_tileserv, and building high-speed geospatial REST API endpoints.

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